

Load-Aware A* Path Planning for Quadruped Robots

Day 24 of 30 · · Week 4: Results Review & Integration · Member 1 — Results Section Input

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1. Nature of Work Done

Day 24 is the capstone review day for Member 1 before final paper assembly begins in Week 5. The four tasks executed today form a closed-loop audit: Methodology v2.0 is re-read against all results data, inconsistencies are identified and classified, amendments are issued where genuinely required, and a cross-section consistency certificate is produced to confirm Sections III, IV, and V are aligned and integration-ready.

Task	Description	Scope	Outcome
Task 1	Final Methodology Review	Re-read all 6 sub-sections of Methodology v2.0 against Day 17–23 outputs	Section-by-section findings log
Task 2	Methodology vs Results Alignment	Map every metric definition in III-F to corresponding results column	Full alignment matrix
Task 3	Amendments Register	Classify findings: CONFIRMED / MINOR UPDATE / CONFLICT; issue corrections	4-item amendment register
Task 4	Cross-Section Consistency Certificate	Final sign-off that Sections III, IV, V are notation- and equation-consistent	Signed certificate — ALL PASS

2. Final Methodology Review — Section-by-Section

Each sub-section of Methodology v2.0 (Sections III-A through III-F) was re-read in full and cross-referenced against: (a) all equation and parameter usage in result computations, (b) the Day 18–20 algorithmic analyses, (c) the Day 22–23 interpretive inputs. Findings are recorded per sub-section below.

2.1 Section III-A — Problem Formulation

Element	Stated in Methodology v2.0	Cross-Check Finding	Status
Grid definition	$R = C = 10$; $G(r,c) \in \{0,1\}$	All 20 scenarios confirmed 10×10 grid with seeded obstacles	CONFIRMED
Path formulation	$\pi = \langle n_0 \dots n_L \rangle$; $n_0 = (0,0)$; $n_L = (9,9)$	Start/goal fixed across all scenarios; path reconstruction verified	CONFIRMED

Element	Stated in Methodology v2.0	Cross-Check Finding	Status
Movement type	4-connected (up/down/left/right only)	get_neighbors() confirmed 4-directional; no diagonal moves in any scenario	CONFIRMED
L definition	Geometric path length in steps	L used as step count in results table; no unit ambiguity detected	CONFIRMED

2.2 Section III-B — Quadruped Robot Model

Element	Stated in v2.0	Cross-Check Finding	Status
Foot positions (Eq. 3)	$p_{FL}=(c-hs, r+hs)$ etc.; $hs = 0.4$ m	Foot span confirmed; support polygon stays within 1.0 m cell; $M(n)$ feasible	CONFIRMED
CoM formula (Eq. 4)	$x_{CoM} = (m^b \cdot x^b + m_{\square} \cdot x_{payload}) / (m^b + m_{\square})$	Applied with $m^b=10$ kg; $m_{\square} \in \{3,5,7,9\}$ kg; x^b =foot centroid	CONFIRMED
Stability Penalty (Eq. 5)	$S(n) = \ x_{CoM}(n) - x_c(n)\ _2$ (L2 norm)	L2 norm confirmed throughout; x_c = support polygon centroid at each cell	CONFIRMED
Stability Margin (Eq. 6)	$M(n) = \min_i d(x_{CoM}, edge_i)$; $M > 0.05$ m threshold	Perpendicular edge distance confirmed; threshold applied consistently in M_{min} column	CONFIRMED
$m^b = 10.0$ kg	Fixed body mass across all scenarios	Constant in all 20 Eq. 4 computations; no scenario deviation	CONFIRMED
$m_{\square} = 3-9$ kg	Payload range; results use $\{3,5,7,9\}$ kg	Four payload levels; 5 scenarios per level in 20-scenario matrix	CONFIRMED

2.3 Section III-C — Standard A* Algorithm

Element	Stated in v2.0	Cross-Check Finding	Status
Cost function (Eq. 7)	$f(n) = g(n) + h(n)$	Baseline results use exactly this; no stability term present in Standard A* runs	CONFIRMED
Heuristic (Eq. 8)	$h(n) = r-r_g + c-c_g $	Admissibility: $h(0,0)=18$ =actual min path on empty grid; verified	CONFIRMED
Step cost (Eq. 9)	$g(n_{next}) = g(n_{cur}) + 1$	Standard A* L values match minimum-step paths in all 20 scenarios	CONFIRMED
Completeness	Always finds path if one exists	All 20 seeds pre-validated solvable; completeness holds	CONFIRMED
Optimality	Returns minimum geometric-length path	$L(Std)$ is minimum for each scenario; no shorter path exists	CONFIRMED

2.4 Section III-D — Load-Aware A* Algorithm

Element	Stated in v2.0	Cross-Check Finding	Status
Modified cost (Eq. 10)	$f(n) = g(n) + \alpha \cdot S(n) + h(n)$; $\alpha = 1.0$	All Load-Aware results computed with $\alpha = 1.0$; confirmed consistent	CONFIRMED
g-update (Eq. 11)	$g(n_{\text{next}}) = g(n_{\text{cur}}) + 1 + \alpha \cdot S(n_{\text{next}})$	S accumulated in g only; h unchanged; admissibility preserved throughout	CONFIRMED
S_map (Eq. 12)	Pre-computed dict; O(1) lookup per node	S_map built once per scenario before search; dT% < 5% confirms this design	CONFIRMED
$\alpha = 0$ backwards compatibility	Reduces exactly to Standard A* when $\alpha = 0$	Regression test required to verify — not yet confirmed in results output	MINOR UPDATE
Novelty N1 (continuous S(n))	Graded penalty; not binary constraint	Results show traversal through moderate-S cells when needed; continuity confirmed	CONFIRMED
$\alpha = 1.0$ rationale	Empirically selected via Day 7 tuning on 10 scenarios	Rationale present in Methodology; needs explicit citation in Section IV prose	MINOR UPDATE

2.5 Section III-E — Implementation Details

Element	Stated in v2.0	Cross-Check Finding	Status
Language / platform	Python 3, Google Colaboratory	All results output consistent with Python environment	CONFIRMED
open_list (heapq)	Min-heap; O(log n) insert/extract	N values consistent with heap-driven search; no anomalous expansion counts	CONFIRMED
closed_set (set())	O(1) membership; prevents re-expansion	No revisited nodes in path traces; closed_set functioning correctly	CONFIRMED
came_from / g_score (dict)	O(1) lookup; parent pointers for path rebuild	Path reconstruction verified; g_score updates match Eq. 11 increments	CONFIRMED
S_map (dict)	Pre-computed; O(1) lookup; Load-Aware only	Single difference from Standard A* per-node cost; no additional overhead loops	CONFIRMED

2.6 Section III-F — Experimental Setup

Element	Stated in v2.0	Cross-Check Finding	Status
20 scenarios	Sufficient for t-test statistical validity	Day 19 matrix: exactly 20 rows; confirmed	CONFIRMED

Element	Stated in v2.0	Cross-Check Finding	Status
Obstacle density	20%–45%	Scenarios span full range; all seeds solvable at stated densities	CONFIRMED
Payload positions	Rows 3–7, Cols 3–7 (interior)	Creates spatial S(n) gradient as intended; all seeds must be verified	MINOR UPDATE
Metrics L, C, N, T	Defined unambiguously with units	All four columns present in results table; units consistent (steps, —, nodes, ms)	CONFIRMED
Metric S_avg	Mean S(n) over path = $(1/L)\sum S(n)$	S_avg column values consistent with this formula across verified scenarios	CONFIRMED
Metric M_min	Min M(n) over path; threshold $M > 0.05$ m	M_min column; threshold applied correctly; safety classification consistent	CONFIRMED
Metric D	True iff paths differ as node sequences	D = True at $m \geq 5$ kg in expected scenarios; definition applied correctly	CONFIRMED
Metrics dL, dT%	Signed/percentage differences	$dL = L_{LA} - L_{Std}$; $dT\% = (T_{LA} - T_{Std})/T_{Std} \times 100$; formulas confirmed	CONFIRMED
$\alpha = 1.0$ in parameter table	Matches Day 7 tuning result	Value consistent; rationale citation pending (AMD-01)	MINOR UPDATE

3. Methodology vs Results Alignment Matrix

This section maps every equation and parameter in Methodology v2.0 to its direct counterpart in the Section V results table. A complete one-to-one correspondence confirms that results are interpretable using only the methodology as written, with no undocumented assumptions or silent deviations.

3.1 Equation-to-Results Traceability

Eq.	Defined In	Appears in Results As	Direct Column	Alignment
Eq. 1	III-A	Defines obstacle cell state; implicit in all scenarios	All scenario grids	CONSISTENT
Eq. 2	III-A	Defines path as ordered sequence; used in D flag and path diagrams	D column; path figures	CONSISTENT
Eq. 3	III-B	Defines support polygon; used in Eq. 5 and 6 computation	Input to S_avg, M_min	CONSISTENT

Eq.	Defined In	Appears in Results As	Direct Column	Alignment
Eq. 4	III-B	CoM position per cell; core of all stability calculations	Input to S_avg, M_min	CONSISTENT
Eq. 5	III-B	S(n) penalty per cell; averaged over path	S_avg column	CONSISTENT
Eq. 6	III-B	M(n) margin per cell; minimum over path	M_min column	CONSISTENT
Eq. 7	III-C	Standard A* evaluation function	C column (Standard A* rows)	CONSISTENT
Eq. 8	III-C	Manhattan heuristic; affects N, T, path shape	Implicit in N, T, L	CONSISTENT
Eq. 9	III-C	Uniform step cost; drives L = minimum steps	L column (Standard A*)	CONSISTENT
Eq. 10	III-D	Load-Aware evaluation function	C column (Load-Aware rows)	CONSISTENT
Eq. 11	III-D	Stability-penalised step cost; drives S_avg and M_min improvement	S_avg, M_min, dL columns	CONSISTENT
Eq. 12	III-D	S_map pre-computation; enables O(1) lookup; explains dT% < 5%	dT% column validation	CONSISTENT

3.2 Parameter-to-Results Traceability

Parameter	v2.0 Value	Usage in Results	Deviation Detected
Grid size R=C	10	10×10 in all 20 scenarios	NO DEVIATION
m ^b body mass	10.0 kg	Constant in all Eq. 4 computations	NO DEVIATION
m□ payload range	{3, 5, 7, 9} kg	5 scenarios per payload level; 20 total	NO DEVIATION
hs foot half-span	0.4 m	Fixed polygon; M_min baseline consistent across all scenarios	NO DEVIATION
Cell size	1.0 m (normalised)	L in steps = L in metres; no conversion error	NO DEVIATION
α stability weight	1.0	Applied in Eq. 11 for all 20 Load-Aware runs	NO DEVIATION
M threshold	M > 0.05 m	Safety classification in M_min column	NO DEVIATION
Scenario count	20	20 rows in Day 19 matrix	NO DEVIATION
Payload position	Rows 3–7, Cols 3–7	Spatial S(n) gradient design	PENDING (AMD-04)

4. Amendments Register — Full Classification

All findings from the Section 2 review are consolidated and classified below. IMPORTANT: Methodology v2.0 is NOT re-issued. All amendments are additions to paper prose sections only. No equations, parameters, or algorithms change.

ID	Severity	Target Section	Amendment Description	Action	Owner
AMD-01	MINOR UPDATE	Section IV prose	Add α -tuning rationale sentence before Novelty Table	Insert one sentence; approved text in Section 2.4	Member 1
AMD-02	MINOR UPDATE	Section V or Appendix	Add $\alpha = 0$ regression test confirmation note	Member 4 runs test; Member 1 writes one sentence	Member 1 + 4
AMD-03	MINOR UPDATE	Section III-B clarification	Clarify x^b = foot centroid = cell centre under symmetric h_s	Add parenthetical to III-B	Member 1
AMD-04	MINOR UPDATE	Section V experimental note	Verify all 20 seeds place payload in rows 3–7, cols 3–7	Spot-check 5 seeds minimum; document result	Member 4

5. Cross-Section Consistency Certificate

This certificate formally confirms that Sections III (Methodology), IV (Algorithm Design), and V (Results) are consistent across notation, equations, parameters, and claims. This certificate is the integration-readiness signal for paper assembly in Week 5.

5.1 Notation Consistency — Symbol Audit

Symbol	Defined in III	Used in IV	Used in V	Consistent
$S(n)$	Stability penalty; L2 norm; metres (Eq. 5)	Additive term in Eq. 10, 11	S_{avg} = mean $S(n)$ over path	CONSISTENT
$M(n)$	Stability margin; perpendicular distance (Eq. 6)	Referenced in safety threshold discussion	M_{min} = min $M(n)$ over path	CONSISTENT
α	Stability weight = 1.0; empirically selected	In Eq. 10, 11; value stated explicitly	Parameter table; all LA runs	CONSISTENT
$g(n)$	Accumulated cost from start	Modified via Eq. 11 ($+\alpha \cdot S$)	Implicit in C column	CONSISTENT

Symb ol	Defined in III	Used in IV	Used in V	Consistent
$h(n)$	Manhattan heuristic (Eq. 8); unchanged in LA	Basis of admissibility argument	Implicit in N, T comparisons	CONSISTENT
D	Path divergence flag (True/False)	Predicted: True when α induces route change	D column; True at $m_{\square} \geq 5$ kg	CONSISTENT
dL	$L_{LA} - L_{Std}$ (signed)	Trade-off efficiency metric	dL column; expected 0–3 steps	CONSISTENT
dT%	$(T_{LA} - T_{Std}) / T_{Std} \times 100$	Overhead < 5% claim ($O(1)$ S_{map})	dT% column; validates $O(1)$ design	CONSISTENT
S_{avg}	Mean $S(n)$ over path cells	Primary effectiveness metric	S_{avg} column; primary result	CONSISTENT
M_{min}	Min $M(n)$ over path cells	Safety-critical metric; threshold 0.05 m	M_{min} column; key result	CONSISTENT

5.2 Equation Consistency — Cross-Section Usage

Consistency Check	Finding	Result
Eq. 4 (CoM): same formula in III-B, IV description, and V computation	Same weighted sum; same parameter values; no variant forms across sections	CONSISTENT
Eq. 5 ($S(n)$): L2 norm used throughout; no Manhattan substitution anywhere	L2 norm confirmed in all references; no section uses a different distance metric	CONSISTENT
Eq. 10/11: $S(n)$ not double-counted in f (g already contains $\alpha \cdot S$)	$f = g_{new} + h$ only; S is embedded in g_{new} via Eq. 11; no duplication found	CONSISTENT
Eq. 11: $S(n_{next})$ used at node arrival, not departure	n_{next} subscript confirmed in all equation references; cost semantics correct	CONSISTENT
Eq. 12: S_{map} covers free cells only ($G(r,c) = 0$)	Obstacle cells excluded; algorithm never expands obstacle nodes; no lookup conflict	CONSISTENT
Admissibility: $h(n) \leq h^*(n)$ given S in $g(n)$ only	$h^*(n)$ = remaining steps; no stability cost in h ; lower bound valid throughout	CONSISTENT

5.3 Major Claims vs Evidence

Claim	Section	Supporting Evidence	Verified
Load-Aware A* achieves lower S_{avg} than Standard A*	IV, VI	$S_{avg}(LA) < S_{avg}(Std)$ across $D=True$ scenarios	CONFIRMED

Claim	Section	Supporting Evidence	Verified
Load-Aware A* achieves higher $M_{\min}$ (safety improvement)	IV, VI	$M_{\min}(\text{LA}) > 0.05 \text{ m}$ vs $M_{\min}(\text{Std}) \leq 0.05 \text{ m}$ in high-payload cases	CONFIRMED
Compute overhead is negligible: $dT\% < 5\%$	IV, V	$O(1)$ S_{map} lookup; $dT\%$ column confirms	CONFIRMED
Algorithm is admissible and combined-cost optimal	IV	Admissibility proof (Day 20); S in g only	CONFIRMED
Payload scalability without re-tuning for $m_{\square} = 3\text{--}9 \text{ kg}$	IV, Conclusion	α fixed; CoM adapts automatically via Eq. 4	CONFIRMED
$dL = 1\text{--}3$ steps is an acceptable stability-cost trade-off	VI	Efficiency frontier analysis; Q1/Q2 quadrant dominance (Day 23)	CONFIRMED
Backwards compatibility: $\alpha = 0 \equiv \text{Standard A}^*$	IV	Mathematical equivalence proven; regression test pending AMD-02	MINOR UPDATE